

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: How humans influence biodiversity - Set 1

Mark scheme

AQA spec 3.1.2.2; The living environment; textbook pp. 20-25

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Food: cod, tuna, swordfish, sharks, turtles; extinct - dodo, great auk, passenger pigeon • Fashion: leopard, snow leopard, tiger, ocelot fur; fur seals; crocodile / alligator leather; kingfisher, parrot, ostrich feathers • Entertainment / pets: parrots, lizards, snakes, tortoises, tropical marine fish (most wild-caught), dolphins / orcas for marine parks, wild-collected air plants • Furniture / ornaments: mahogany, teak, ramin timber; ebony and ivory piano keys; turtle shell, shark teeth, coral and shells as souvenirs • Traditional medicines: tiger (bones, claws), rhinoceros horn, seahorses	5
01.2	<i>Any 3 from:</i> • Threat to humans: sharks, venomous snakes, crocodiles • Disease vectors: malaria mosquitoes, tsetse flies • Predators of livestock: wolves, puma, birds of prey, herons, seals • Agricultural / forestry pests and competitors: rabbits, deer, wood-boring beetles, squirrels, beavers	3
01.3	• High demand and prices make illegal trade / poaching profitable; laws are hard to enforce across borders • Belief in the medicine persists despite little scientific evidence; species become rarer and more valuable	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 20-21; AQA spec 3.1.2.2.	

Question	Marking guidance	Marks
02.1	• Drainage of wetlands / lowering water tables by groundwater abstraction dries out wetlands; HEP schemes alter river levels flooding sandbank nests of turtles / birds; species with narrow tolerance (specialists) affected first; sundews lost when drained soil becomes aerobic / nitrogen-rich and taller competitors invade • Hot water discharges and organic wastes (sewage) reduce dissolved oxygen, killing aerobic organisms such as fish and insect larvae • Acid mine drainage and acid rain gases lower pH, denaturing proteins of exposed tissues (fish eggs, gills) and dissolving calcium exoskeletons (crayfish)	6
02.2	• Reduces light penetration so submerged plants cannot photosynthesise • Blocks the gills of filter feeders such as bivalve molluscs; sediment smothers the bed	2
02.3	• Rare species often have very specialised habitat requirements / narrow ranges of tolerance • A small change moves conditions outside their range while generalists survive; small populations cannot recover	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 22-23; AQA spec 3.1.2.2.	
03.1	• As sea otters declined (1960-1980) sea urchins increased (to 350) and kelp area fell (to 25) • As otters recovered after 1990 urchins declined and kelp recovered • Otter and kelp trends are positively correlated; urchins negatively correlated with both	3
03.2	• Sea otters are predators of sea urchins so fewer otters allowed the urchin population to rise • Sea urchins graze kelp so the kelp forest was over-grazed and destroyed • Kelp forests are habitat / food for many species including otters (keystone species / trophic cascade); otter recovery restored the balance	3
03.3	<i>Any 2 from:</i> • Over-fishing of sandeels reduced puffins • Over-collection of turtle eggs reduced turtles so jellyfish (their food) increased; over-fishing of sardines / herring (competitors) also increased jellyfish • Loss of pollinating bees (pesticides, wild flower loss) reduces plant reproduction; loss of elephants / large herbivores reduces seed dispersal of trees	2
03.4	• They control abiotic features: beaver dams create ponds colonised by aquatic species; elephants create clearings and keep water holes open • Species that depend on these features (light, water, minerals) cannot survive in their absence	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 23-25; AQA spec 3.1.2.2.	
04.1	• Island communities evolved from few colonising species in isolation, often without mammal predators or large herbivores • So indigenous species lack adaptations (eg ground-nesting birds, flightlessness) to survive introduced cats, rats, pigs, goats	2

Question	Marking guidance	Marks
04.2	• Competitor: grey squirrel (North America) out-competes red squirrel - digests acorns, larger, wins nest sites; rhododendron shades ground and releases toxins preventing native regrowth • Predator: American mink (escaped fur farms) reduced water vole range; cane toads in Australia; cats / rats / pigs on islands eating ground-nesting birds; Nile perch in Lake Victoria ate cichlids to extinction • Pathogen: squirrel pox virus carried by grey squirrels kills red squirrels; crayfish plague carried by signal crayfish kills white-clawed crayfish; Dutch elm disease, ash dieback, sudden oak death imported on plants / soil	6
04.3	• Closely related introduced species (sika deer; domestic cats) cross-breed producing fertile hybrids • The natural gene pool is changed / diluted with genes it would not naturally contain, so the pure indigenous species may disappear	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 24-25; AQA spec 3.1.2.2.	
05.1	• B	1
05.2	<i>Any 4 from:</i> • Deforestation • Ploughing of grassland / expansion of farmland • Reservoir creation / flooding • Mineral extraction, especially open-cast mining • Urban expansion	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 20-26; AQA spec 3.1.2.2.	